

Debian 12 Server Installation Guide

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1. Introduction

This guide explains how to install a Debian 12 server without a graphical environment on a virtual machine using Qemu/KVM. It covers setting up the system and then installing Apache, PostgreSQL, PHP, and PhpPgAdmin.

Each major step includes a list of mandatory screenshots to support clear, visual documentation in your final report.

2. Pre-Installation Preparation

2.1. Create a Working Directory and Prepare Your Notes

Create a Directory:

Create a new directory called “S2.03”, it will be used to store all your notes, command logs and screenshots.

```
$ mkdir ~/S2.03
```

pro tip: While writing commands, don't forget to use a text editor to record them and document its effect.

2.2. Getting and Verifying the ISO Image

Download or use the pre-downloaded Debian 12 netinst ISO x86_64 from:

<https://cdimage.debian.org/cdimage/release/current/amd64/iso-cd/>

Locate the ISO:

The file is provided in /usr/local/images-ISO/. Check its contents with:

```
$ ls /usr/local/images-ISO/
```

Verify Integrity:

Compare the SHA-512 checksum of the ISO:

```
$ sha512sum NOM_FICHER.iso
```

```
meharyn@iut2-dg027-d15:~$ sha512sum /usr/local/images-ISO/debian-12.10.0-amd64-netinst.iso  
cb089def0684fd93c9c2fbe45fd16ecc809c949a6fd0c91ee199faefe7d4b82b64658a264a13109d59f1a40ac3080b  
e2f7bd3d8bf3e9cdf509add6d72576a79b /usr/local/images-ISO/debian-12.10.0-amd64-netinst.iso
```

exemple of usage of the sha512sum command

Now, compare the returned value with:

<https://cdimage.debian.org/cdimage/release/current/amd64/iso-cd/SHA512SUMS>

If it is the same, you can go to the next step

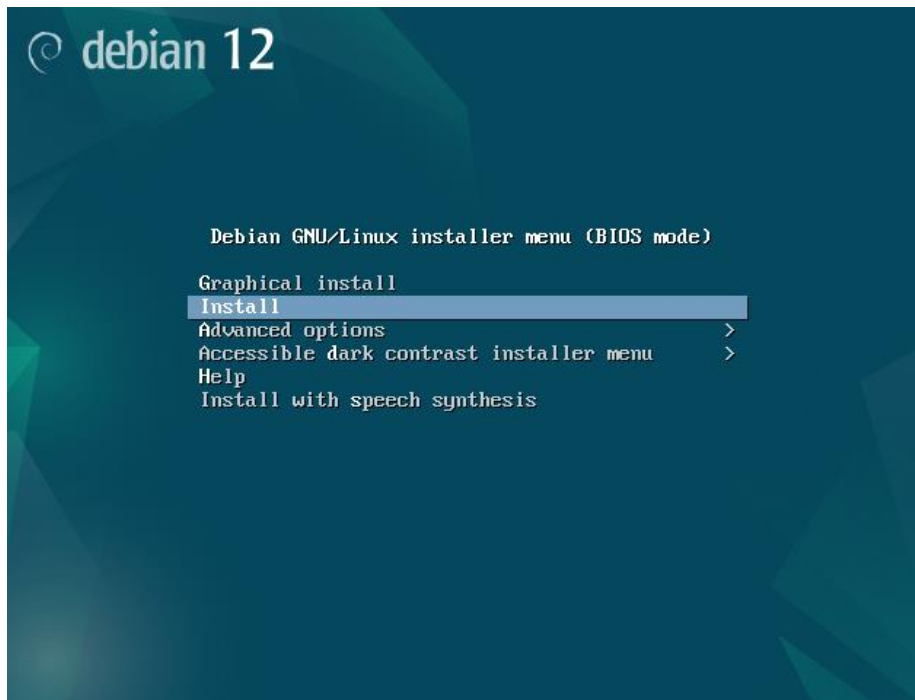
3. Installing Debian 12 via Qemu/KVM

3.1. Booting from the ISO

Launch the Installer:

Run the following command to start the virtual machine on the ISO image that you just installed

```
$ S2.03-lance-installation
```



This window should now open on QEMU

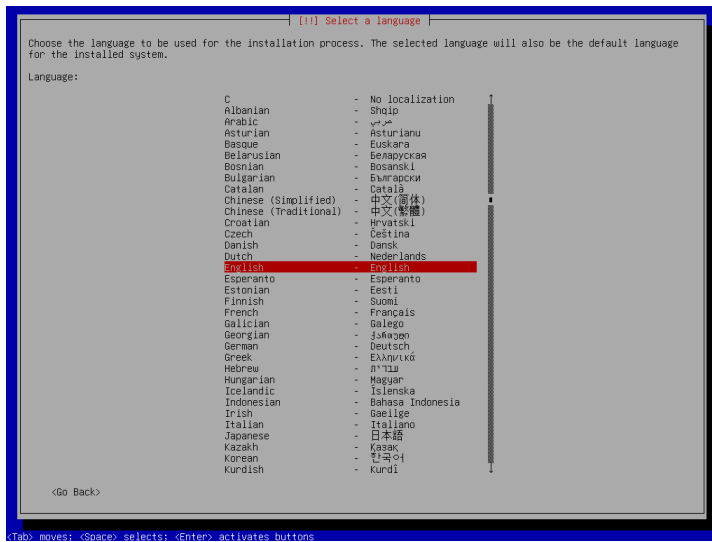
3.1. Installation Procedure

You are now about to setup your Debian distribution

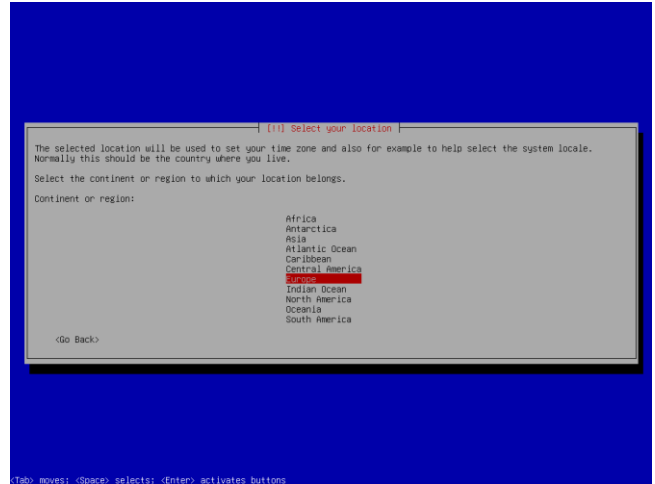
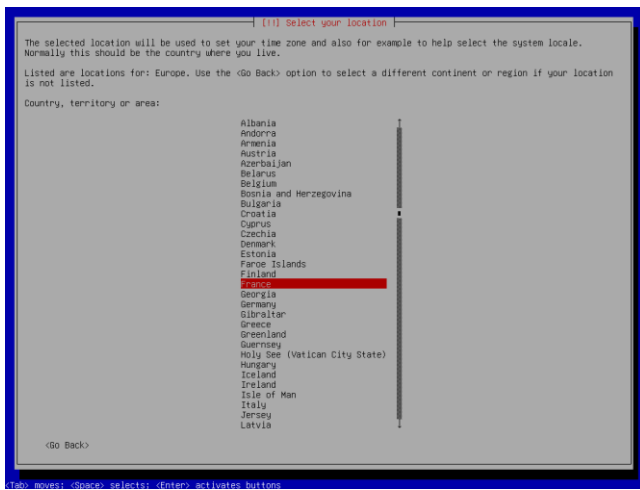
Follow each Debian installer step:

Language and Location:

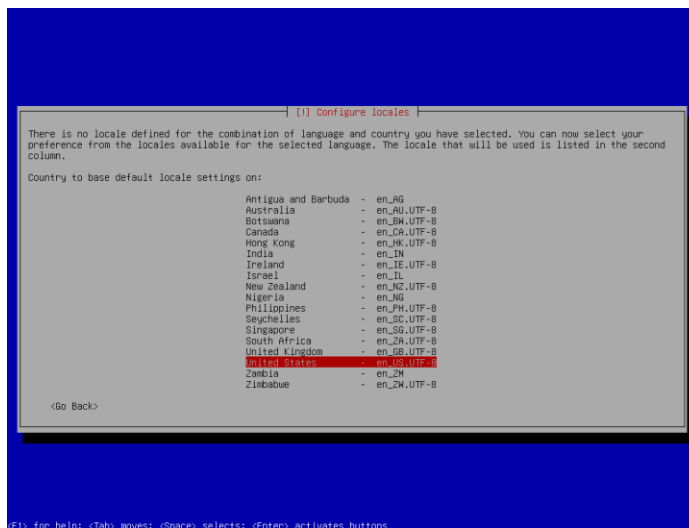
⑩ **Language:** Choose **English**.



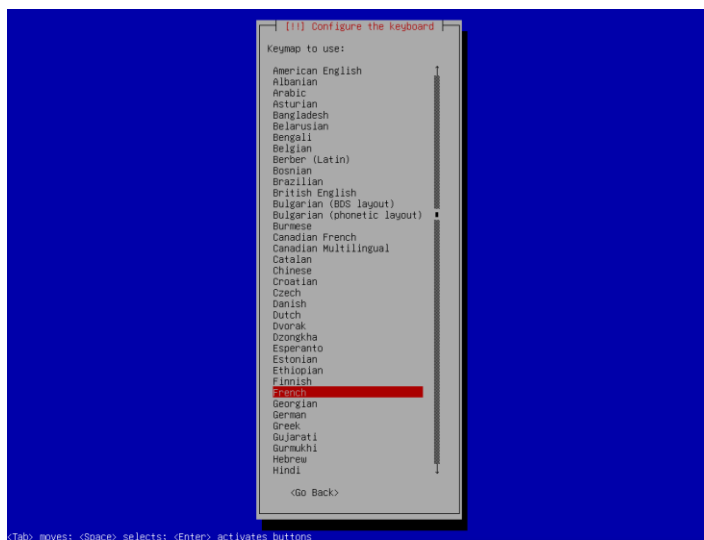
⑩ Location: Select other/Europe/France.



⑩ Locales: Choose United States, en_US.UTF-8.

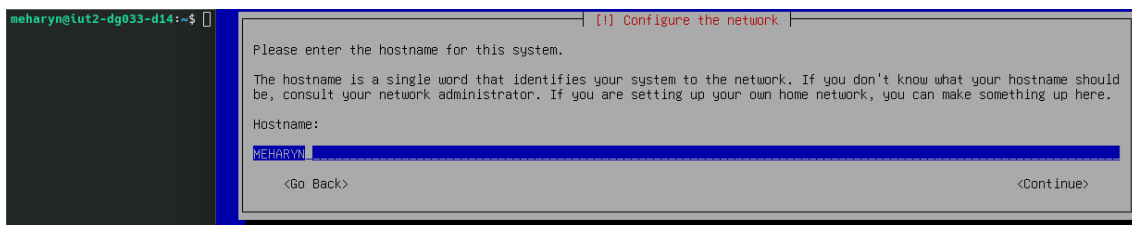


⑩ Keyboard: Select French.



⑩ Hostname and User Setup:

⑩ Hostname: Enter server-⟨YOUR_UGA_LOGIN⟩.



⑩ Set Root Password:

Use a simple password (e.g., “root”) and check “Show Password.”

⑩ User Account:

⑩ Full Name: (e.g., “Jean Toto”)

⑩ Username: Your UGA login

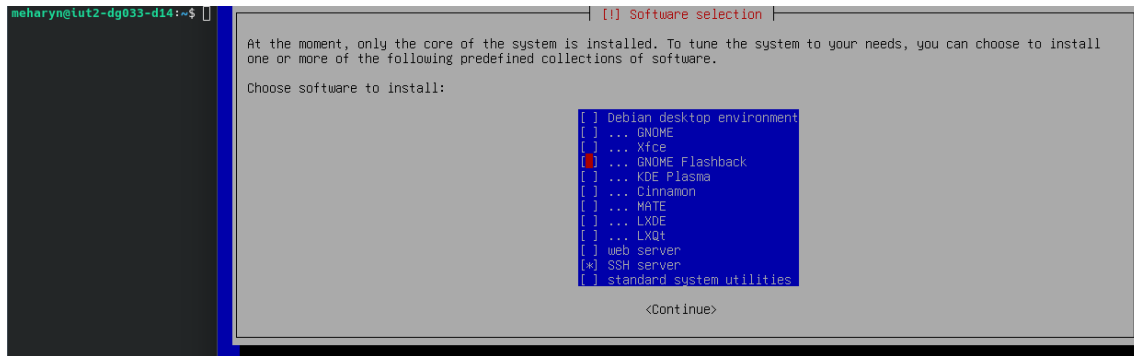
⑩ Password: A simple password (e.g., “etu”) with “Show Password” checked.

⑩ Disk Partitioning:

- ⑩ Choose **Guided** – use entire disk.
- ⑩ Select **All files in one partition**.
- ⑩ Confirm with **Yes**.

⑩ Software Selection:

- ⑩ Ensure that **Debian desktop** is unchecked and **SSH server** is checked.



⑩ GRUB Installation:

- ⑩ Confirm GRUB installation on **/dev/sda**.
- Capture 8:** Screenshot showing the GRUB installation prompt.

3.2. Completing Installation and Shutting Down

After installation, the VM will restart.
Shut it down properly by logging in as root and executing:

```
# systemctl poweroff
```

```
root@server-meharyn:/home/meharyn# systemctl poweroff
```

pro tip: always make sure you close the virtual machine this way, which is the only proper way, else, you might encounter data loss, damaged VM files or file system corruption